

# FUNDAMENTALS OF ELECTRICAL ENGINEERING

1. Electrical engineering is a branch of engineering that deals with the study and application of electricity, electronics, and electromagnetism.
2. The field of electrical engineering covers a wide range of subfields including electronics, digital computers, power engineering, telecommunications, control systems, radio-frequency engineering, and signal processing.
3. Electrical engineers design, develop, test, and supervise the manufacturing of electrical equipment, such as electric motors, radar and navigation systems, communications systems, and power generation equipment.
4. The basic concepts of electrical engineering include electric charge, electric current, electric potential, and electric power.
5. Electric charge is a fundamental property of matter that results from the presence or absence of electrons. Electric current is the flow of electric charge, and electric potential is the difference in electric potential energy per unit charge between two points.
6. Electric power is the rate at which electric energy is transferred by an electric circuit, and is typically measured in watts.
7. Electrical engineering also involves the study of electromagnetic fields and their interactions with matter, including the principles of electromagnetism, electromagnetic induction, and electromagnetic waves.
8. Electrical engineers use a variety of tools and techniques to design and analyze electrical systems, including circuit analysis, computer-aided design (CAD), and simulation software.
9. Electrical engineering is a constantly evolving field, with new technologies and applications being developed all the time. Some of the current trends in electrical engineering include renewable energy, smart grids, and the Internet of Things (IoT).
10. To become an electrical engineer, one typically needs to complete a bachelor's degree in electrical engineering or a related field. Many electrical engineers also choose to pursue advanced degrees, such as a master's or PhD, to specialize in a particular area of the field.

## Unit 1 - DC Circuits

1. **Introduction to DC Circuits:** A DC circuit is an electrical circuit that consists of any combination of constant voltage sources, constant current sources, and resistors. In this type of circuit, the current is constant, meaning it doesn't change with time.
2. **Ohm's Law:** Ohm's law states that the current through a conductor between two points is directly proportional to the voltage across the two points. Mathematically, it is represented as  $V = IR$ , where  $V$  is the voltage,  $I$  is the current, and  $R$  is the resistance.

3. **Series and Parallel Circuits:** In a series circuit, the current is the same through all components, while the voltage is divided among the components. In a parallel circuit, the voltage is the same across all components, while the current is divided among the components.
4. **Kirchhoff's Laws:** Kirchhoff's Current Law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero. Kirchhoff's Voltage Law (KVL) states that the algebraic sum of all voltages around a closed loop (or mesh) is zero.
5. **Power in DC Circuits:** The power in a DC circuit is the product of the voltage and the current,  $P = VI$ . The unit of power is the watt (W).
6. **Resistors in Series and Parallel:** The total resistance of resistors in series is the sum of their individual resistances. The total resistance of resistors in parallel is the reciprocal of the sum of the reciprocals of their individual resistances.
7. **Capacitors in Series and Parallel:** The total capacitance of capacitors in series is the reciprocal of the sum of the reciprocals of their individual capacitances. The total capacitance of capacitors in parallel is the sum of their individual capacitances.
8. **DC Circuit Analysis:** DC circuit analysis involves finding the voltage, current, and power in each component of a DC circuit. This can be done using techniques such as Ohm's law, Kirchhoff's laws, and nodal analysis.
9. **Thevenin's and Norton's Theorems:** Thevenin's theorem states that any linear, bilateral, two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source in series with a resistor. Norton's theorem states that any linear, bilateral, two-terminal circuit can be replaced by an equivalent circuit consisting of a current source in parallel with a resistor.
10. **Maximum Power Transfer Theorem:** The maximum power transfer theorem states that the maximum power is transferred from a source to a load when the load resistance is equal to the internal resistance of the source.

### Electrical circuit elements (R, L and C)

An electrical circuit is made up of various elements, each with its own unique properties and functions. The three most common elements found in electrical circuits are resistors (R), inductors (L), and capacitors (C).

1. **Resistors (R):** A resistor is a passive two-terminal electrical component that implements electrical resistance as a circuit element. In electronic circuits, resistors are used to reduce current flow, adjust signal levels, to divide voltages, bias active elements, and terminate transmission lines, among other uses.

2. **Inductors (L):** An inductor, also known as a coil or reactor, is a passive two-terminal electrical component that stores energy in a magnetic field when electric current flows through it. Inductors are used in a variety of electrical applications, including in power supplies, filters, transformers, and motors.
3. **Capacitors (C):** A capacitor is a passive two-terminal electrical component that stores electrical energy in an electric field. The effect of a capacitor is known as capacitance. Capacitors are widely used in electronic circuits for blocking direct current while allowing alternating current to pass, in filter networks, for smoothing the output of power supplies, in the resonant circuits that tune radios to particular frequencies, in electric power transmission systems for stabilizing voltage and power flow, and for many other purposes.

These three elements, along with other components such as voltage and current sources, switches, and diodes, can be combined in various ways to form complex electrical circuits. Understanding the properties and functions of these elements is essential for analyzing and designing electrical circuits.

### **Concept of active and passive elements for the notes of the Unit 1 - DC Circuits in the subject of FUNDAMENTALS OF ELECTRICAL ENGINEERING**

- **Active elements** are components that are capable of generating or amplifying electrical energy. They are capable of controlling the flow of electrons and can provide gain or amplification to a circuit. Examples of active elements include transistors, operational amplifiers, and voltage-controlled oscillators.
- **Passive elements**, on the other hand, are components that do not generate or amplify electrical energy. They are incapable of controlling the flow of electrons and do not provide gain or amplification to a circuit. Examples of passive elements include resistors, capacitors, and inductors.
- In a DC circuit, active elements are typically used to provide power to the circuit, while passive elements are used to control the flow of current and voltage within the circuit.
- Active and passive elements play important roles in the design and operation of DC circuits. Understanding the differences between these two types of elements is essential for anyone studying or working with electrical engineering.

### **Voltage and Current Sources**

Voltage and current sources are fundamental components of electrical circuits. They are used to provide power to the circuit and to control the flow of current.

1. **Voltage sources** provide a constant voltage across their terminals, regardless of the current flowing through them. Examples of voltage sources include batteries and power supplies.
2. **Current sources** provide a constant current through their terminals, regardless of the voltage across them. Examples of current sources include photovoltaic cells and constant current generators.
3. Voltage and current sources can be either independent or dependent. Independent sources provide a fixed voltage or current, while dependent sources provide a voltage or current that is controlled by another variable in the circuit.
4. In DC circuits, voltage and current sources are typically represented by symbols in circuit diagrams. The symbol for a voltage source is a circle with a “+” sign inside, while the symbol for a current source is a circle with an arrow pointing in the direction of the current flow.
5. Voltage and current sources are essential for the analysis and design of DC circuits. They allow us to control the flow of current and to provide power to the circuit components.

### Concept of Linearity

Linearity is a mathematical property that is used to describe the behavior of a system or a component. In the context of DC circuits, linearity refers to the relationship between the voltage and current in a circuit element.

A circuit element is said to be linear if its voltage-current relationship follows Ohm’s law, which states that the current through a conductor between two points is directly proportional to the voltage across the two points. This means that if the voltage across a linear circuit element is doubled, the current through it will also double.

In a linear circuit, the output is directly proportional to the input. This means that if the input voltage or current is increased, the output voltage or current will also increase by the same factor. Linear circuits are easier to analyze and design because their behavior can be predicted using simple mathematical equations.

Some common examples of linear circuit elements include resistors, capacitors, and inductors. These elements obey Ohm’s law and their behavior can be described using linear equations.

In contrast, non-linear circuit elements do not follow Ohm’s law and their behavior cannot be described using simple linear equations. Some examples of non-linear circuit elements include diodes, transistors, and some types of voltage and current sources.

In summary, linearity is an important concept in the study of DC circuits. It refers to the relationship between the voltage and current in a circuit element

and is used to describe the behavior of linear and non-linear circuit elements. Understanding linearity is essential for the analysis and design of DC circuits.

### Unilateral and Bilateral Elements

Unilateral and bilateral elements are terms used to describe the behavior of electrical components in a circuit. These terms are important to understand when studying DC circuits in the subject of Fundamentals of Electrical Engineering.

- **Unilateral Elements:** A unilateral element is an electrical component that allows current to flow in only one direction. This means that the voltage and current characteristics of the element are dependent on the direction of current flow. Examples of unilateral elements include diodes and transistors.
- **Bilateral Elements:** A bilateral element is an electrical component that allows current to flow in both directions. This means that the voltage and current characteristics of the element are independent of the direction of current flow. Examples of bilateral elements include resistors, capacitors, and inductors.

It is important to note that the behavior of unilateral and bilateral elements can change depending on the type of circuit they are used in. For example, a diode may behave as a bilateral element in an AC circuit, but as a unilateral element in a DC circuit.

In summary, unilateral and bilateral elements are important concepts to understand when studying DC circuits in the subject of Fundamentals of Electrical Engineering. Unilateral elements allow current to flow in only one direction, while bilateral elements allow current to flow in both directions. The behavior of these elements can change depending on the type of circuit they are used in.

### Kirchhoff's laws for the notes of the Unit 1 - DC Circuits in the subject of FUNDAMENTALS OF ELECTRICAL ENGINEERING

Kirchhoff's laws are two equalities that deal with the current and potential difference (commonly known as voltage) in the lumped element model of electrical circuits. They were first described in 1845 by Gustav Kirchhoff. These laws are widely used in electrical engineering.

1. **Kirchhoff's Current Law (KCL):** This law is also known as Kirchhoff's first law, Kirchhoff's point rule, or Kirchhoff's junction rule. This law states that, for any node (junction) in an electrical circuit, the sum of currents flowing into that node is equal to the sum of currents flowing out of that node. Mathematically, the algebraic sum of currents in a network of conductors meeting at a point is zero.
2. **Kirchhoff's Voltage Law (KVL):** This law is also known as Kirchhoff's second law, Kirchhoff's loop (or mesh) rule, and Kirchhoff's second rule.

This law states that the directed sum of the potential differences (voltages) around any closed network is zero. Mathematically, the algebraic sum of the products of the resistances of the conductors and the currents in them in a closed loop is equal to the total emf available in that loop.

These laws can be applied to any circuit, whether it is a simple circuit with a few components or a complex circuit with many components and branches. They are fundamental tools for the analysis of circuits and are widely used in the study of electrical engineering.

### **Mesh and Nodal Methods of Analysis**

Mesh and nodal analysis are two fundamental methods for analyzing DC circuits. These methods are used to determine the current and voltage in each branch of a circuit.

#### **Mesh Analysis**

Mesh analysis is a method used to solve for the unknown currents in a circuit. It is based on Kirchhoff's Voltage Law (KVL), which states that the sum of the voltage drops around any closed loop in a circuit must be equal to zero.

To perform mesh analysis, the following steps are followed:

1. Identify all the meshes in the circuit. A mesh is a loop that does not contain any other loops within it.
2. Assign a current to each mesh, flowing in a clockwise direction.
3. Write KVL equations for each mesh, expressing the voltage drops in terms of the mesh currents.
4. Solve the system of equations to find the values of the mesh currents.

#### **Nodal Analysis**

Nodal analysis is a method used to solve for the unknown voltages in a circuit. It is based on Kirchhoff's Current Law (KCL), which states that the sum of the currents entering a node must be equal to the sum of the currents leaving that node.

To perform nodal analysis, the following steps are followed:

1. Identify all the nodes in the circuit. A node is a point where two or more circuit elements are connected.
2. Select a reference node, usually the one with the most connections, and assign it a voltage of zero.
3. Assign a voltage to each of the remaining nodes.
4. Write KCL equations for each node, expressing the currents in terms of the node voltages.
5. Solve the system of equations to find the values of the node voltages.

Both mesh and nodal analysis are powerful tools for analyzing DC circuits and can be used to solve complex circuits with multiple sources and loads. They are commonly used in circuit analysis and design.

## Unit 2 - Steady State Analysis of Single Phase AC Circuits

1. **Introduction:** Steady state analysis of single phase AC circuits involves the calculation of current, voltage, and power in circuits that are powered by a single phase AC source.
2. **Sinusoidal Steady State:** In sinusoidal steady state, all the circuit variables such as current, voltage, and power are sinusoidal functions of time with the same frequency as the source.
3. **Phasor Representation:** Phasors are complex numbers that represent the magnitude and phase of sinusoidal signals. They are used to simplify the analysis of AC circuits.
4. **Impedance and Admittance:** Impedance is the opposition to the flow of current in an AC circuit. It is a complex quantity that takes into account both resistance and reactance. Admittance is the reciprocal of impedance.
5. **Power in AC Circuits:** In AC circuits, power is the product of voltage and current. It can be calculated using the phasor representation of voltage and current.
6. **Power Factor:** Power factor is the ratio of real power to apparent power. It is an important parameter in the analysis of AC circuits as it affects the efficiency of power transmission.
7. **Series and Parallel AC Circuits:** The analysis of series and parallel AC circuits is similar to that of DC circuits. However, the use of phasors and complex impedance makes the calculations easier.
8. **Resonance in AC Circuits:** Resonance occurs in AC circuits when the reactance of the circuit is zero. At resonance, the circuit behaves as a purely resistive circuit and the current is maximum.

### Representation of Sinusoidal waveforms – Average and effective values

## Unit 2 - Steady State Analysis of Single Phase AC Circuits

### FUNDAMENTALS OF ELECTRICAL ENGINEERING

1. A sinusoidal waveform is a mathematical curve that describes a smooth repetitive oscillation.
2. It is named after the function sine, of which it is the graph.
3. The average value of a sinusoidal waveform is the average of all the instantaneous values during one cycle.
4. The average value of a sinusoidal waveform over one complete cycle is zero.
5. The effective value of a sinusoidal waveform is the value of a direct current that would produce the same average power in a resistive load.

6. The effective value is also known as the root mean square (RMS) value.
7. The RMS value of a sinusoidal waveform is equal to its peak value divided by the square root of two.
8. The RMS value is used to calculate the power in an AC circuit.
9. The average and effective values are important for the analysis of single phase AC circuits in steady state.

### Form and Peak Factors

- In the steady-state analysis of single-phase AC circuits, sinusoidal waveforms are represented using average and effective values, form and peak factors.
- The peak-to-peak value of an alternating quantity is the value from its positive peak to its negative peak.
- The average value of an alternating quantity is the arithmetic mean of all the values over a complete cycle.
- The form factor is the ratio of the effective value to the average value of an alternating quantity.
- The peak factor is the ratio of the maximum value to the effective value of an alternating quantity.

### Analysis of single phase AC Circuits consisting R-L-C combination (Series and Parallel)

### Unit 2 - Steady State Analysis of Single Phase AC Circuits

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1. **Introduction:** In single phase AC circuits, the analysis of circuits consisting of resistors (R), inductors (L), and capacitors (C) in series or parallel combinations is important to understand the behavior of the circuit.
2. **Series R-L-C Circuit:** In a series R-L-C circuit, the current is the same through all the components. The voltage across each component is different and is determined by the impedance of the component. The total impedance of the circuit is the sum of the individual impedances.
3. **Parallel R-L-C Circuit:** In a parallel R-L-C circuit, the voltage is the same across all the components. The current through each component is different and is determined by the admittance of the component. The total admittance of the circuit is the sum of the individual admittances.
4. **Impedance and Admittance:** Impedance is the opposition to the flow of current in an AC circuit. It is a complex quantity and is represented as  $Z = R + jX$ , where R is the resistance and X is the reactance. Admittance is the reciprocal of impedance and is represented as  $Y = 1/Z$ .
5. **Phasor Diagrams:** Phasor diagrams are used to represent the magnitude and phase of the voltage and current in an AC circuit. They are useful



in visualizing the relationships between the different components in the circuit.

6. **Power Factor:** Power factor is the ratio of the real power to the apparent power in an AC circuit. It is an important parameter in the analysis of AC circuits as it affects the efficiency of the circuit.
7. **Resonance:** Resonance occurs in an R-L-C circuit when the inductive reactance and the capacitive reactance are equal in magnitude. At resonance, the circuit behaves as a purely resistive circuit and the power factor is unity.
8. **Conclusion:** The analysis of single phase AC circuits consisting of R-L-C combinations in series or parallel is important in understanding the behavior of the circuit. The use of phasor diagrams, impedance and admittance calculations, and power factor analysis can provide valuable insights into the operation of the circuit. Resonance is a special condition that can occur in R-L-C circuits and has important implications for the circuit's behavior.

### Apparent, Active & Reactive Power

In the study of AC circuits, power is often expressed in terms of three different quantities: apparent power, active power, and reactive power.

1. **Apparent Power** is the product of the RMS values of voltage and current. It is measured in volt-amperes (VA) and is represented by the letter 'S'.
2. **Active Power** is the average power consumed by a circuit over a complete cycle. It is measured in watts (W) and is represented by the letter 'P'.
3. **Reactive Power** is the power that is stored and returned to the source in each cycle. It is measured in volt-amperes reactive (VAR) and is represented by the letter 'Q'.

In a purely resistive circuit, the apparent power is equal to the active power, and there is no reactive power. However, in circuits with inductance and capacitance, the apparent power is greater than the active power, and there is a non-zero reactive power.

The relationship between apparent, active, and reactive power can be represented by a power triangle, where the apparent power is the hypotenuse, the active power is the adjacent side, and the reactive power is the opposite side.

The power factor of a circuit is defined as the ratio of the active power to the apparent power. It is a measure of how effectively the circuit is using the power supplied to it. A power factor of 1 indicates that all the power supplied to the circuit is being used effectively, while a power factor of less than 1 indicates that some of the power is being wasted.

In summary, apparent power, active power, and reactive power are important concepts in the study of AC circuits. They help us understand how power is

consumed and stored in different types of circuits, and how effectively the power is being used. Understanding these concepts is essential for the steady-state analysis of single-phase AC circuits.

### **Power factor for the notes of the Unit 2 - Steady State Analysis of Single Phase AC Circuits in the subject of FUNDAMENTALS OF ELECTRICAL ENGINEERING**

- Power factor is the ratio of real power to apparent power in an AC circuit.
- It is a measure of how effectively the current is being converted into useful work output.
- The power factor can range from 0 to 1, with 1 being the most efficient.
- In a purely resistive circuit, the phase angle is  $= 0$ , and the power factor is unity ( $\cos = 1$ ), which means that the current is in phase with the input (supply) voltage.
- Power factor can be improved by adding power factor correction capacitors to the AC circuit.
- Power factor is important in AC circuits because it affects the efficiency of power transmission and can result in increased energy costs if it is not close to 1.

### **Concept of Resonance in Series & Parallel Circuits**

Resonance is a phenomenon that occurs in both series and parallel AC circuits. It is characterized by a large increase in current or voltage at a particular frequency, known as the resonant frequency.

#### **Series Resonance**

In a series resonant circuit, the resonant frequency is determined by the values of the inductance (L) and capacitance (C) in the circuit. At the resonant frequency, the inductive reactance (XL) and capacitive reactance (XC) are equal in magnitude but opposite in sign, resulting in a net reactance of zero. This means that the circuit behaves as a purely resistive circuit at the resonant frequency, allowing maximum current to flow.

The resonant frequency of a series resonant circuit can be calculated using the formula:

$$f = 1 / (2 * \sqrt{LC})$$

where f is the resonant frequency, L is the inductance, and C is the capacitance.

#### **Parallel Resonance**

In a parallel resonant circuit, the resonant frequency is also determined by the values of the inductance and capacitance in the circuit. At the resonant frequency, the inductive susceptance (BL) and capacitive susceptance (BC) are

equal in magnitude but opposite in sign, resulting in a net susceptance of zero. This means that the circuit behaves as an open circuit at the resonant frequency, allowing minimum current to flow.

The resonant frequency of a parallel resonant circuit can be calculated using the same formula as for a series resonant circuit:

$$f = 1 / (2 * \sqrt{LC})$$

where  $f$  is the resonant frequency,  $L$  is the inductance, and  $C$  is the capacitance.

### Applications of Resonance

Resonance has many practical applications in electrical engineering. For example, it is used in the design of filters, oscillators, and tuned circuits. In a filter circuit, resonance can be used to selectively pass or block certain frequencies. In an oscillator circuit, resonance can be used to generate a stable frequency. In a tuned circuit, resonance can be used to maximize the transfer of energy between two circuits.

### Bandwidth and Quality Factor

- The **bandwidth** of a series circuit is defined as the range of frequencies in which the amplitude of the current is equal to or greater than  $(1 / 2 = 2 / 2)$  times its maximum amplitude.
- The **quality factor** or **Q factor** is a dimensionless parameter that describes how underdamped an oscillator or resonator is, and characterizes a resonator's bandwidth relative to its center frequency.
- The bandwidth can be calculated using the formula  $BW = f_c / Q$  where  $f_c$  is the resonant frequency and  $Q$  is the quality factor.
- A high  $Q$  resonant circuit has a narrow bandwidth as compared to a low  $Q$ .
- Bandwidth is measured between the 0.707 current amplitude points. The 0.707 current points correspond to the half power points since  $P = I^2 R$ ,  $(0.707)^2 = (0.5)$ .

### Three phase balanced circuits

Three phase balanced circuits are an important topic in the study of steady state analysis of single phase AC circuits, which is covered in Unit 2 of the subject Fundamentals of Electrical Engineering.

1. The electrical system can be of two types: single-phase and three-phase. The single-phase system has only one phase wire and one return wire, and is used for low power transmission .
2. It is always better to solve balanced three-phase circuits on the basis of each phase. When the three-phase supply voltage is given without reference to the line or phase value, then it is the line voltage which is taken into consideration .

3. The analysis of three phase balanced circuits is no different from the analysis of AC systems in general. The relation between voltages, currents and power in delta-connected and star-connected systems has already been discussed in earlier sections .
4. Circuits or systems in which the AC sources operate at the same frequency but different phases are known as polyphase. A three-phase system is produced by a generator (alternator), whose cross-sectional view is shown in Figure 2(a) .

### **Voltage and Current Relations in Star and Delta Connections**

#### **Star Connection:**

- In a star connection, one end of each winding is connected to a common point, called the neutral point.
- The phase voltage is the voltage across each winding.
- The line voltage is the voltage between any two lines.
- The line voltage is equal to the square root of 3 times the phase voltage.
- The phase current is the current flowing through each winding.
- The line current is equal to the phase current.

#### **Delta Connection:**

- In a delta connection, the windings are connected end-to-end to form a closed loop.
- The phase voltage is the voltage across each winding.
- The line voltage is equal to the phase voltage.
- The phase current is the current flowing through each winding.
- The line current is equal to the square root of 3 times the phase current.

## **Unit 3 - Transformers**

Transformers are electrical devices that transfer electrical energy between two or more circuits through electromagnetic induction. They are used to increase or decrease the voltage of an alternating current (AC) power supply.

Some key points to remember about transformers are:

1. Transformers operate on the principle of electromagnetic induction.
2. They consist of two or more coils of wire wrapped around a magnetic core.
3. The primary coil is connected to the input power supply, while the secondary coil is connected to the output load.
4. When an AC voltage is applied to the primary coil, it creates a magnetic field that induces a voltage in the secondary coil.
5. The ratio of the number of turns in the primary coil to the number of turns in the secondary coil determines the voltage transformation ratio.
6. Transformers can be used to step-up or step-down the voltage of an AC power supply.

7. They are widely used in power transmission and distribution systems, as well as in electronic devices such as power adapters and chargers.

### **Magnetic Circuits**

Magnetic circuits are used to create a path for magnetic flux. They are similar to electric circuits in that they have a source of magnetomotive force (MMF), which is analogous to electromotive force (EMF) in an electric circuit. The MMF is produced by a current flowing through a coil of wire, which creates a magnetic field.

Magnetic circuits also have a magnetic reluctance, which is analogous to resistance in an electric circuit. The reluctance is determined by the properties of the materials in the circuit, such as their permeability, and the geometry of the circuit.

Magnetic circuits are used in many applications, including transformers, motors, and generators. In a transformer, the primary and secondary windings are wound around a magnetic core, which provides a low-reluctance path for the magnetic flux. This allows for efficient transfer of energy from the primary to the secondary winding.

Some key points to remember about magnetic circuits are: - Magnetic circuits have a source of MMF, which is produced by a current flowing through a coil of wire. - Magnetic circuits have a magnetic reluctance, which is determined by the properties of the materials in the circuit and the geometry of the circuit. - Magnetic circuits are used in many applications, including transformers, motors, and generators. - In a transformer, the primary and secondary windings are wound around a magnetic core, which provides a low-reluctance path for the magnetic flux.

### **Ideal and Practical Transformer**

A transformer is a device that transfers electrical energy from one circuit to another through electromagnetic induction. It is used to change the voltage level of an alternating current (AC) power supply.

#### **Ideal Transformer**

An ideal transformer is an imaginary transformer that has the following characteristics:

1. It has no losses, meaning that the power input is equal to the power output.
2. It has no leakage flux, meaning that all the magnetic flux is confined to the core and links both the primary and secondary windings.
3. It has an infinite magnetizing inductance, meaning that the magnetizing current is zero.

In an ideal transformer, the voltage ratio is equal to the turns ratio, and the current ratio is the inverse of the turns ratio.

### **Practical Transformer**

A practical transformer, on the other hand, is not ideal and has some losses and imperfections. These include:

1. Copper losses: These are the power losses due to the resistance of the windings.
2. Iron losses: These are the power losses due to the hysteresis and eddy currents in the core.
3. Leakage flux: This is the magnetic flux that does not link both the primary and secondary windings.

In a practical transformer, the voltage ratio is not exactly equal to the turns ratio due to the voltage drop across the winding resistance and the leakage reactance. The current ratio is also not exactly the inverse of the turns ratio due to the magnetizing current.

Despite these imperfections, practical transformers can still achieve high efficiency and are widely used in power systems, electrical machines, and electronic devices.

### **Equivalent Circuit for the Notes of the Unit 3 - Transformers in the Subject of Fundamentals of Electrical Engineering**

An equivalent circuit is a simplified representation of a complex circuit that retains the essential characteristics of the original circuit. In the case of a transformer, the equivalent circuit is used to model the behavior of the transformer under different operating conditions.

The equivalent circuit of a transformer consists of the following components:

1. Primary winding resistance ( $R_p$ ): This represents the resistance of the primary winding of the transformer.
2. Secondary winding resistance ( $R_s$ ): This represents the resistance of the secondary winding of the transformer.
3. Primary leakage reactance ( $X_p$ ): This represents the leakage reactance of the primary winding of the transformer.
4. Secondary leakage reactance ( $X_s$ ): This represents the leakage reactance of the secondary winding of the transformer.
5. Magnetizing reactance ( $X_m$ ): This represents the magnetizing reactance of the transformer core.
6. Core loss resistance ( $R_c$ ): This represents the core loss resistance of the transformer core.

The equivalent circuit can be used to calculate the voltage, current, and power in the primary and secondary windings of the transformer. It can also be used

to determine the efficiency and regulation of the transformer.

### Losses in Transformers

Transformers are used to transfer electrical energy from one circuit to another through electromagnetic induction. However, during this process, some energy is lost in the form of heat. These losses are known as transformer losses. There are two main types of losses in transformers: core losses and copper losses.

1. **Core losses:** Core losses, also known as iron losses, occur in the magnetic core of the transformer. These losses are caused by two main factors: hysteresis and eddy currents. Hysteresis losses occur due to the repeated magnetization and demagnetization of the core, while eddy current losses occur due to the circulation of currents within the core. Core losses are constant and do not depend on the load on the transformer.
2. **Copper losses:** Copper losses, also known as winding losses, occur in the windings of the transformer. These losses are caused by the resistance of the windings and are proportional to the square of the current flowing through them. Copper losses increase with an increase in the load on the transformer.

In addition to these two main types of losses, there are also other minor losses such as stray losses and dielectric losses. These losses are usually small and can be neglected for most practical purposes.

To minimize losses in transformers, the core is usually made of high-quality magnetic materials with low hysteresis and eddy current losses. The windings are also carefully designed to minimize resistance and hence reduce copper losses.

In summary, losses in transformers are an inevitable part of the energy transfer process. However, careful design and selection of materials can help minimize these losses and improve the efficiency of the transformer.

### Regulation and Efficiency

Transformers are used to transfer electrical energy from one circuit to another through electromagnetic induction. The efficiency and regulation of a transformer are important factors to consider when designing and operating these devices.

#### Regulation

Regulation refers to the ability of a transformer to maintain a constant secondary voltage despite changes in the load. It is defined as the percentage change in secondary voltage from no-load to full-load conditions. A transformer with good regulation will have a small percentage change in secondary voltage, indicating that the secondary voltage remains relatively constant despite changes in the load.

## Efficiency

Efficiency is a measure of how well a transformer converts electrical energy from the primary circuit to the secondary circuit. It is defined as the ratio of the output power to the input power, expressed as a percentage. A transformer with high efficiency will have a small amount of energy loss during the conversion process.

Factors that affect the efficiency of a transformer include the resistance of the windings, the core losses (hysteresis and eddy current losses), and the leakage flux. To improve the efficiency of a transformer, these losses must be minimized.

In summary, regulation and efficiency are important factors to consider when designing and operating transformers. Good regulation ensures that the secondary voltage remains constant despite changes in the load, while high efficiency ensures that the energy conversion process is efficient with minimal losses.

## Unit 4 - Electrical machines

Electrical machines are devices that convert electrical energy into mechanical energy or vice versa. There are two main types of electrical machines: motors and generators.

1. **Motors** convert electrical energy into mechanical energy. They work on the principle of electromagnetic induction, where a current-carrying conductor experiences a force when placed in a magnetic field. This force is used to produce rotational motion. There are several types of motors, including DC motors, AC motors, and stepper motors.
2. **Generators** convert mechanical energy into electrical energy. They also work on the principle of electromagnetic induction, where a conductor moving in a magnetic field induces an electromotive force (EMF) in the conductor. This EMF drives a current through an external circuit. There are several types of generators, including DC generators and AC generators.

Electrical machines are used in a wide range of applications, including transportation, power generation, and industrial processes. They are essential components of modern society and play a crucial role in our daily lives.

### DC Machines

DC machines are a type of electrical machine that converts mechanical energy into direct current (DC) electrical energy or vice versa. DC machines are commonly used in applications that require a constant and controllable speed, such as in elevators, electric vehicles, and industrial machinery.

Some key points to note about DC machines are:



1. DC machines consist of two main components: the stator and the rotor. The stator is the stationary part of the machine, while the rotor is the rotating part.
2. The stator contains the field windings, which produce a magnetic field when energized. The rotor contains the armature windings, which interact with the magnetic field to produce torque.
3. The commutator is a key component of DC machines. It is responsible for switching the current in the armature windings, allowing the machine to produce a continuous and controllable output.
4. The speed of a DC machine can be controlled by varying the voltage applied to the field windings or by varying the resistance in the armature circuit.
5. DC machines can be classified into two main types: DC generators and DC motors. DC generators convert mechanical energy into electrical energy, while DC motors convert electrical energy into mechanical energy.
6. DC machines can also be classified based on their field excitation method. There are three main types: separately excited, shunt-wound, and series-wound.
7. DC machines have several advantages, including high starting torque, good speed control, and high efficiency. However, they also have some disadvantages, such as the need for regular maintenance and the production of sparks and noise due to the commutator.

### Principle & Construction of Electrical Machines

- Electrical machines are devices that convert mechanical energy to electrical energy and vice versa.
- All rotating electric machines have two basic parts in common. The first is the rotating part known as the rotor and the second is the stationary part called the stator. These parts are made up of highly permeable magnetic material such as silicon steel.
- The working principle and operation of a DC machine is based on an effect when a current-carrying conductor coils laying in a magnetic field, the magnetic field produces a mechanical force on it known as torque which rotates the conductor coils in the magnetic field.
- The construction is the same for DC motor and DC generator.
- Multi-machine systems, brushless motors, and switched reluctance motors are covered, as well as constant flux and constant current operation of induction motors.

### Types of Electrical Machines

Electrical machines can be broadly classified into two types:

1. **Static Machines:** These refer to stationary machines such as transformers, which may be single or three-phase transformers .

2. **Dynamic Machines:** These refer to both DC and AC machines. DC machines include both DC motors and generators, while AC machines include synchronous and asynchronous machines .

Some of the main types of electrical machines are:

- **Transformer:** In a transformer, both the input and output are electrical power .
- **Generator:** An electric generator is a device that converts mechanical energy to electrical energy .
- **Motor:** An electric motor converts electricity to mechanical power .

### **EMF Equation of Generator and Torque Equation of Motor**

#### **EMF Equation of Generator**

The EMF equation of a generator is used to determine the generated EMF in a generator. The equation is given as follows:

$$EMF = (\Phi * Z * N * P) / (60 * A)$$

Where: - EMF is the generated EMF in volts -  $\Phi$  is the flux per pole in webers - Z is the total number of armature conductors - N is the speed of the armature in revolutions per minute (RPM) - P is the number of poles - A is the number of parallel paths in the armature

#### **Torque Equation of Motor**

The torque equation of a motor is used to determine the torque produced by a motor. The equation is given as follows:

$$T = (\Phi * I_a * Z * P) / (2 * A)$$

Where: - T is the torque produced by the motor in newton-meters (Nm) -  $\Phi$  is the flux per pole in webers -  $I_a$  is the armature current in amperes - Z is the total number of armature conductors - P is the number of poles - A is the number of parallel paths in the armature - 2 is a constant value, approximately equal to 6.28

These equations are important for understanding the operation of electrical machines and are commonly used in the study of the fundamentals of electrical engineering. They can be used to calculate the generated EMF and torque produced by a generator or motor, respectively. It is important to note that the values of the variables in these equations must be known in order to accurately calculate the EMF or torque.

#### Unit 4 - Electrical Machines: Applications of DC Motors (Simple Numerical Problems)

DC motors are widely used in various applications due to their versatility and efficiency. Some of the common applications of DC motors include:

1. **Electric vehicles:** DC motors are used in electric vehicles to convert electrical energy into mechanical energy to drive the wheels. The high starting torque and speed control of DC motors make them suitable for this application.
2. **Industrial machinery:** DC motors are used in various industrial machinery such as lathes, drills, and milling machines. The speed control and high starting torque of DC motors make them suitable for these applications.
3. **Household appliances:** DC motors are used in various household appliances such as vacuum cleaners, washing machines, and fans. The efficiency and speed control of DC motors make them suitable for these applications.
4. **Robotics:** DC motors are used in robotics to provide motion to the robot. The speed control and high starting torque of DC motors make them suitable for this application.

Simple numerical problems involving DC motors can be solved using the basic equations governing the operation of DC motors. These equations include the torque equation, the speed equation, and the power equation.

Example problem:

A DC motor has an armature resistance of 0.5 ohms and is connected to a 240V supply. The motor draws a current of 50A when running at a speed of 1500 rpm. Calculate the torque developed by the motor.

Solution:

The torque developed by the motor can be calculated using the torque equation:  
 $T = k * I_a * \phi$

Where T is the torque, k is a constant,  $I_a$  is the armature current, and  $\phi$  is the flux.

The value of k can be calculated using the speed equation:  $N = (V - I_a * R_a) / (k * \phi)$

Where N is the speed, V is the supply voltage,  $I_a$  is the armature current,  $R_a$  is the armature resistance, and  $\phi$  is the flux.

Substituting the given values into the speed equation, we get:

$$1500 = (240 - 50 * 0.5) / (k * \phi)$$

Solving for k, we get  $k = 0.24 / \phi$

Substituting this value of  $k$  into the torque equation, we get:

$$T = (0.24 / \phi) * 50 * \phi$$

Solving for  $T$ , we get  $T = 12 \text{ Nm}$

Therefore, the torque developed by the motor is 12 Nm.

### Three Phase Induction Motor

A three-phase induction motor is an AC electric motor that operates on three-phase power. It is one of the most common types of electric motors used in industry, due to its simplicity, reliability, and low cost.

Some key points to note about three-phase induction motors are:

1. The stator of a three-phase induction motor is made up of a series of windings that are energized by a three-phase AC power source. This creates a rotating magnetic field within the motor.
2. The rotor of a three-phase induction motor is typically made of a series of conductive bars that are shorted together at either end by end rings. This is known as a “squirrel cage” rotor.
3. When the rotating magnetic field created by the stator windings passes over the rotor bars, it induces a current in the bars. This current creates its own magnetic field, which interacts with the stator’s magnetic field to produce torque.
4. The speed of a three-phase induction motor is determined by the frequency of the AC power source and the number of poles in the motor. The synchronous speed of the motor can be calculated using the formula:  $N_s = 120f/P$ , where  $N_s$  is the synchronous speed in RPM,  $f$  is the frequency of the AC power source in Hz, and  $P$  is the number of poles in the motor.
5. The actual speed of a three-phase induction motor will be slightly less than its synchronous speed, due to slip. Slip is the difference between the synchronous speed and the actual speed of the motor, expressed as a percentage of the synchronous speed.
6. Three-phase induction motors are available in a wide range of sizes and power ratings, from fractional horsepower to several thousand horsepower.
7. Three-phase induction motors are commonly used in applications such as pumps, fans, compressors, and conveyor systems.

### Principle & Construction for the notes of the Unit 4 - Electrical machines in the subject of FUNDAMENTALS OF ELECTRICAL ENGINEERING

1. Electrical machines are devices that convert mechanical energy to electrical energy and vice versa.

2. The working, construction, and principle of DC and AC machines are covered in these notes.
3. The notes cover the concept of control of speed, generation of Torque, various losses, efficiency, etc. of the electromechanical machines such as motors and generators in the real world.
4. The book provides an elaborate and systematic analysis of the working principle, applications and construction of each electrical machine.
5. In addition to circuit responses under steady state conditions, the book contains the chapters on dynamic responses of networks and analysis of a three-phase circuit.
6. The course teaches the principles and analysis of electromechanical systems.
7. Students will develop analytical techniques for predicting device and system interaction characteristics as well as learn to design major classes of electric machines.
8. Construction of 3-phase Induction Motors:
  - It has two main parts:
    1. Stator: the stationary part.
    2. Rotor: the rotating part.
  - There is a small air gap between the rotor and stator (0.4 mm to 4 mm) depend on the power of the motor.

### Types of Electrical Machines

Electrical machines are devices that convert electrical energy into mechanical energy or vice versa. There are three main types of electrical machines:

1. **Generators:** These machines convert mechanical energy into electrical energy. They work on the principle of electromagnetic induction, where a conductor moving in a magnetic field induces an electromotive force (EMF) in the conductor.
2. **Motors:** These machines convert electrical energy into mechanical energy. They work on the principle of electromagnetic force, where a current-carrying conductor experiences a force when placed in a magnetic field.
3. **Transformers:** These machines transfer electrical energy from one circuit to another without changing its frequency. They work on the principle of mutual induction, where a changing magnetic field in one coil induces an EMF in another coil.

### Slip-torque characteristics for the notes of the Unit 4 - Electrical machines in the subject of FUNDAMENTALS OF ELECTRICAL ENGINEERING

- The slip-torque characteristic of an induction motor describes the relationship between the slip of the motor and the torque produced by the motor.

- Slip is defined as the difference between the synchronous speed of the rotor and the actual speed of the rotor, expressed as a percentage of the synchronous speed.
- The torque produced by an induction motor is directly proportional to the slip of the motor, up to a certain point.
- As the slip increases, the torque produced by the motor also increases, until it reaches a maximum value known as the breakdown torque.
- Beyond the breakdown torque, the torque produced by the motor decreases as the slip increases.
- The slip-torque characteristic of an induction motor is an important factor in determining the performance of the motor in various applications.
- By understanding the slip-torque characteristic of an induction motor, it is possible to select the appropriate motor for a given application and to optimize its performance.

#### Applications (Numerical problems related to slip only)

1. **Slip Calculation:** Slip is an important parameter in the operation of an induction motor. It is defined as the difference between the synchronous speed and the rotor speed, expressed as a fraction of the synchronous speed. The formula for calculating slip is:  $\text{Slip} = (N_s - N_r) / N_s$ , where  $N_s$  is the synchronous speed and  $N_r$  is the rotor speed.

2. **Example Problem:** A 4-pole, 50 Hz induction motor is operating at a speed of 1425 RPM. Calculate the slip of the motor.

**Solution:** The synchronous speed of the motor can be calculated using the formula  $N_s = 120f / P$ , where  $f$  is the frequency and  $P$  is the number of poles. Substituting the values, we get  $N_s = 120 * 50 / 4 = 1500$  RPM.

The slip of the motor can now be calculated using the formula  $\text{Slip} = (N_s - N_r) / N_s$ . Substituting the values, we get  $\text{Slip} = (1500 - 1425) / 1500 = 0.05$ .

Therefore, the slip of the motor is 0.05 or 5%.

3. **Significance of Slip:** Slip is an important parameter in the operation of an induction motor. It determines the torque produced by the motor and also affects its efficiency. A low slip indicates that the motor is operating close to its synchronous speed and is producing high torque. A high slip, on the other hand, indicates that the motor is operating at a lower speed and is producing less torque. It is important to maintain the slip within an acceptable range to ensure the efficient operation of the motor.

#### Single Phase Induction Motor

A single-phase induction motor is an electric motor that operates on single-phase AC power. It is one of the most common types of electric motors used in household appliances, such as fans, washing machines, and refrigerators.

1. **Construction:** A single-phase induction motor consists of a stator and a rotor. The stator is the stationary part of the motor and contains the windings. The rotor is the rotating part of the motor and is usually made of a laminated iron core with slots for the conductors.
2. **Working Principle:** When a single-phase AC supply is applied to the stator windings, a magnetic field is produced. This magnetic field rotates around the stator at synchronous speed. The rotating magnetic field induces a current in the rotor conductors, which in turn produces a magnetic field around the rotor. The interaction between the stator and rotor magnetic fields causes the rotor to rotate.
3. **Types of Single Phase Induction Motors:** There are several types of single-phase induction motors, including split-phase, capacitor-start, capacitor-run, shaded-pole, and repulsion-start motors. Each type has its own unique characteristics and is suitable for specific applications.
4. **Applications:** Single-phase induction motors are widely used in household appliances, such as fans, washing machines, and refrigerators. They are also used in small industrial and commercial applications, such as pumps, compressors, and conveyors.
5. **Advantages:** Single-phase induction motors are simple in construction, reliable, and relatively inexpensive. They are also easy to maintain and have a long service life.
6. **Disadvantages:** Single-phase induction motors have lower starting torque and efficiency compared to three-phase induction motors. They also have a lower power factor and can be more difficult to control.

**Principle of operation and introduction to methods of starting for the notes of the Unit 4 - Electrical machines in the subject of FUNDAMENTALS OF ELECTRICAL ENGINEERING**

1. Electrical machines are devices that convert electrical energy into mechanical energy or vice versa.
2. The principle of operation of electrical machines is based on the interaction between magnetic fields and electric currents.
3. The two main types of electrical machines are motors and generators.
4. Motors convert electrical energy into mechanical energy, while generators convert mechanical energy into electrical energy.
5. There are several methods of starting electrical machines, including direct-on-line starting, star-delta starting, and soft starting.
6. Direct-on-line starting involves applying the full voltage to the motor windings, which can result in high starting currents and mechanical stress on the machine.
7. Star-delta starting reduces the starting current by connecting the motor windings in a star configuration during starting, and then switching to a

delta configuration for normal operation.

8. Soft starting gradually increases the voltage applied to the motor windings, reducing the starting current and mechanical stress on the machine.
9. The choice of starting method depends on factors such as the size and type of the machine, the load characteristics, and the power supply conditions.

#### **Applications for the notes of the Unit 4 - Electrical machines in the subject of FUNDAMENTALS OF ELECTRICAL ENGINEERING**

1. Electrical machines are used in a wide range of applications, including power generation, transmission, and distribution.
2. They are used in various industries, such as manufacturing, mining, and transportation, to convert electrical energy into mechanical energy or vice versa.
3. Electrical machines are also used in household appliances, such as washing machines, refrigerators, and air conditioners, to perform various tasks.
4. In the field of transportation, electrical machines are used in electric vehicles, trains, and aircraft to provide propulsion.
5. Electrical machines are also used in renewable energy systems, such as wind turbines and solar panels, to generate electricity.
6. In the field of robotics and automation, electrical machines are used to provide motion and control to various systems.
7. Electrical machines are also used in medical equipment, such as MRI machines and ventilators, to provide essential functions.

#### **Three Phase Synchronous Machines**

Three-phase synchronous machines are a type of electrical machine that operates at a constant speed, determined by the frequency of the alternating current (AC) power supply. These machines are commonly used as generators in power plants and as motors in industrial applications.

Some key points to note about three-phase synchronous machines include:

1. The rotor of a synchronous machine is magnetized by direct current (DC) and rotates at a constant speed, determined by the frequency of the AC power supply.
2. The stator of a synchronous machine contains three-phase windings, which produce a rotating magnetic field when energized by an AC power supply.
3. The interaction between the rotating magnetic field of the stator and the magnetic field of the rotor produces torque, which causes the rotor to rotate at a constant speed.
4. Synchronous machines can operate as either motors or generators, depending on the direction of power flow.
5. In generator mode, mechanical energy is converted into electrical energy, while in motor mode, electrical energy is converted into mechanical energy.



6. Synchronous machines are highly efficient and are commonly used in applications where constant speed is required.

### **Principle of operation of alternator and synchronous motor**

#### **Alternator**

- An alternator is an electrical machine that converts mechanical energy into electrical energy in the form of alternating current (AC).
- The basic principle of operation of an alternator is electromagnetic induction, which states that an electromotive force (EMF) is induced in a conductor when it is subjected to a changing magnetic field.
- In an alternator, the mechanical energy is used to rotate a magnetic field (rotor) inside a set of stationary windings (stator), which induces an alternating EMF in the stator windings.
- The frequency of the generated AC voltage is determined by the speed of rotation of the rotor and the number of magnetic poles on the rotor.

#### **Synchronous Motor**

- A synchronous motor is an AC motor that operates at a constant speed, determined by the frequency of the AC power supply and the number of magnetic poles on the rotor.
- The basic principle of operation of a synchronous motor is similar to that of an alternator, with the roles of the rotor and stator reversed.
- In a synchronous motor, the stator windings are energized with an AC voltage, which creates a rotating magnetic field. The rotor, which is magnetized, is attracted to the rotating magnetic field and starts to rotate at the same speed as the magnetic field.
- The rotor of a synchronous motor can be magnetized either by a direct current (DC) supplied through slip rings or by permanent magnets.

### **Unit 4 - Electrical Machines: Applications**

1. All electrical machines use applications of electro-magnetic principles where electric currents create magnetic fields, which either attract or repel each other. This is the basis of all electric motors, whether they operate on alternating current (AC), direct current (DC) or are universal motors that operate on both .
2. The use of DC machines, i.e., DC generators and motors are very limited. They are mainly used in supplying excitation of small and medium-range alternators. The Industrial Applications of DC Machines are in Electrolytic Processes, Welding processes and Variable speed motor drives .
3. Electric motors are found in applications as diverse as industrial fans, blowers and pumps, machine tools, household appliances, power tools,

and disk drives. They may be powered by direct current or by alternating current which leads to the two main classifications: AC motors and DC motors .

4. Rotating electric machines include DC electric machines – DC motors and DC generators, Synchronous machines – Alternators and Synchronous motors, and Induction motors or Asynchronous machines .
5. Electric machines are essential systems in electric vehicles and are widely used in other applications. In particular, permanent magnet direct current (PMDC) motors have been extensively employed in industrial applications such as electric vehicles .

## **Unit 5 - Electrical Installations**

1. Electrical installations refer to the fixed electrical equipment that is supplied through the electrical wiring system of a building or structure.
2. Electrical installations include equipment such as lighting fixtures, power outlets, switches, and circuit breakers.
3. Electrical installations must comply with local and national electrical codes and regulations to ensure safety and proper functioning.
4. Electrical installations are typically performed by licensed electricians who have the necessary training and expertise to install and maintain electrical systems.
5. Proper electrical installations are essential for the safe and efficient operation of electrical equipment and appliances.
6. Electrical installations can vary in complexity, from simple installations in residential homes to complex installations in commercial and industrial buildings.
7. Regular maintenance and inspection of electrical installations is important to ensure their continued safe and efficient operation.
8. Upgrades and improvements to electrical installations can improve energy efficiency and reduce energy costs.

### **Introduction of Switch Fuse Unit (SFU)**

A Switch Fuse Unit (SFU) is a type of electrical switchgear that combines the functions of a switch and a fuse. It is used to protect electrical installations from overcurrent and short circuit faults. The SFU is typically installed in a distribution board or panel board and is used to control and protect individual circuits.

Some key points to note about SFUs are: - SFUs are available in different current ratings to suit the requirements of the circuit being protected. - The switch component of the SFU allows for manual isolation of the circuit. - The fuse component of the SFU provides overcurrent and short circuit protection. - In the event of a fault, the fuse will blow and interrupt the flow of current, protecting

the circuit and preventing damage to equipment. - SFUs are commonly used in industrial and commercial electrical installations.

### **MCB**

MCB stands for Miniature Circuit Breaker. It is an automatically operated electrical switch designed to protect an electrical circuit from damage caused by excess current from an overload or short circuit. Its basic function is to interrupt current flow after a fault is detected.

- MCBs are used in low voltage electrical networks.
- They are designed to trip quickly on overloads and short circuits.
- MCBs are available in different current ratings and tripping characteristics.
- They are resettable and can be easily turned back on after tripping.
- MCBs are commonly used in residential, commercial, and industrial applications.

### **ELCB**

- ELCB stands for Earth Leakage Circuit Breaker.
- It is a safety device used in electrical installations with high Earth impedance to prevent electric shocks.
- It detects small stray voltages on the metal enclosures of electrical equipment and interrupts the circuit if a dangerous voltage is detected.
- ELCB is capable of detecting the slight amount of current in a faulty metal electrical equipment.
- If any current flows to the ground wire, it is because of electrical faults and the supply of current to that appliance must be stopped, so ELCBs are used to protect the person who is in contact with the electrical device.
- The main function of an ELCB is to prevent shock while electrical installations through high Earth impedance.
- ELCBs do not offer protection against overloading or short circuit, therefore, they must be used in series with an MCB (miniature circuit breaker).

### **MCCB**

MCCB stands for Molded Case Circuit Breaker. It is a type of electrical protection device that is used to protect the electrical circuit from excessive current, which can cause overload or short circuit. With a current rating of up to 2500A, MCCBs can be used for a wide range of voltages and frequencies with adjustable trip settings.

An MCCB is made from the following main parts: - Arc chute - Contacts - Operating mechanism - Terminal Connector - Thermal Trip Unit - Magnetic Trip Unit

MCCB is used to protect the low voltage distribution system. It is available in rating up to 2500 Amps and 1.1 kV.

Selectivity is an important aspect of MCCBs. Only the faulty part of the installation should be isolated. All the other switching and protective devices connected to the system should remain operative. Selectivity reduces the duration of a fault and limits its possible damaging effect only to a part of the installation.

### **ACB**

An Air Circuit Breaker (ACB) is a type of circuit breaker that uses air as the insulation medium to protect an electrical circuit from damage caused by excess current from an overload or short circuit. It is typically used in low voltage applications, such as in buildings and industrial plants.

Some key points to remember about ACBs are:

1. ACBs are used to protect electrical circuits from damage caused by excess current.
2. They use air as the insulation medium.
3. They are typically used in low voltage applications.
4. They can be found in buildings and industrial plants.

### **Types of Wires for the notes of the Unit 5 - Electrical Installations in the subject of FUNDAMENTALS OF ELECTRICAL ENGINEERING**

1. **Solid Wire:** A solid wire is a single conductor wire that is usually made of copper or aluminum. It is commonly used in residential and commercial wiring for power and lighting circuits.
2. **Stranded Wire:** A stranded wire is made up of multiple small gauge wires that are twisted together to form a larger conductor. Stranded wires are more flexible than solid wires and are commonly used in applications where the wire needs to be moved or bent frequently.
3. **Braided Wire:** A braided wire is made up of multiple strands of wire that are woven together to form a flexible conductor. Braided wires are commonly used in applications where high flexibility and durability are required.
4. **Coaxial Cable:** A coaxial cable is a type of cable that has an inner conductor surrounded by a tubular insulating layer, which is then surrounded by a tubular conducting shield. Coaxial cables are commonly used in telecommunications and data transmission applications.
5. **Ribbon Cable:** A ribbon cable is a flat, wide cable that consists of multiple insulated wires running parallel to each other. Ribbon cables are

commonly used in computer and electronic systems to connect multiple components.

6. **Twisted Pair Cable:** A twisted pair cable consists of two insulated wires twisted together. Twisted pair cables are commonly used in telecommunications and data transmission applications to reduce electromagnetic interference.
7. **Fiber Optic Cable:** A fiber optic cable is a type of cable that uses light to transmit data. Fiber optic cables are commonly used in telecommunications and data transmission applications due to their high bandwidth and long-distance capabilities.

### Cables and Bus-bars

Cables and bus-bars are used in electrical installations to distribute electrical power. Here are some key points to note:

1. **Cables** are insulated conductors used to transmit electrical power from one point to another. They are available in different sizes and types, depending on the voltage level and current-carrying capacity required.
2. **Bus-bars** are thick strips of copper or aluminum used to distribute electrical power within switchgear, distribution boards, and substations. They are designed to carry large currents and are typically uninsulated.
3. The selection of cables and bus-bars depends on several factors, including the voltage level, current-carrying capacity, and the environment in which they will be installed.
4. Cables and bus-bars must be properly sized and installed to ensure the safe and efficient operation of the electrical installation.
5. Proper maintenance of cables and bus-bars is essential to prevent failures and ensure the reliability of the electrical installation.

### Fundamentals of Earthing and Lightning Protection

1. **Lightning Protection System:** A lightning protection system intercepts or diverts lightning and provides a certain path for conducting the surges safely to the ground by adequate down conductors to grounding electrodes. Thus, it helps prevent disastrous events like fires, injuries, and deaths.
2. **Lightning Protection Grounding:** Lightning protection grounding helps protect structures and equipment from direct strikes. Overhead ground wires and surge arresters, connected to ground, can limit dangerous system overvoltages to safe values. Fundamentally, grounding an electronic system is the same as grounding any electrical system.

3. **Lightning Rod:** In a lightning protection system, the lightning rod is a single component of the system. The lightning rod requires a connection to earth to give a building protection. Lightning rods come in many different forms, including hollow, solid, pointed or rounded. All lightning rods are made of conductive materials, such as copper and aluminum.
4. **Common Methods for Earthing and Lightning Protection:** It's a matter of providing an alternative path for the lightning current to flow to ground and ensure that: 1. Lightning current flowing in the lightning protection system (LPS) does not induce dangerous currents in any parallel metallic systems near the LPS.
5. **Importance of Earthing and Lightning Protection:** Earthing and Lightning protection are important for everyone that uses electrical equipment and that includes the large majority of the world. Earthing, Lightning Arrestors and Surge Protection Devices are the leading devices that protect devices against current shock.

### Types of Batteries

Batteries can be broadly divided into two major types: 1. Primary Cell / Primary battery 2. Secondary Cell / Secondary battery

Based on the application of the battery, they can be classified again. They are:  
 - Household Batteries: These are the types of batteries which are more likely to be known to the common man.

The following are the types of batteries that are explained with their uses: 1. Lead-acid batteries 2. Nickel-cadmium batteries (Ni-Cd) 3. Nickel-metal hybrid batteries (Ni-MH) 4. Lithium-ion batteries (Li-ion) 5. Alkaline batteries 6. Zinc-carbon batteries 7. Coin cell batteries 8. Zinc-air cells 9. Sealed lead-acid batteries.